

Institute of Technology

Name of Program: MCA

Course Type: Core

Semester: I, **Academic Year:** 2026-27, **Term:** ODD

Course Code & Name	:	6CS103CC26-Operating Systems
-------------------------------	----------	-------------------------------------

Sr. No	Practical Title	Hours	CO
1	Demonstrate various Unix commands and prepare a document with screenshots of the running of different commands. (Appendix – 1)	4	3
2	Write a shell script to perform the following operation: • Implement the basic calculator. (Using decision making and case control structure). • Generate a series of numbers multiplied by 2 up to 50 numbers. • Generate all the combinations of 1, 2, and 3.	2	3
3	Write a shell script to perform the following operation: • Keep accepting lines of text and write the text into a data file until the user inputs "end". The script should count the number of lines input and display it. • Compare the contents of two files. Instruction: Use diff, cmp, and comm to compare the content of the file and print the output with different options.	2	3
4	Write a shell script for implementing the following directory management functions: • Navigation using Absolute/ Relative Pathnames • Listing Directories • Creating Directories • Modifying Directories	4	3
5	Create a file employees.txt with the following fields Emp_ID Emp_Name Designation Department Salary Age Write a shell script to perform basic DBMS functions: • Insert data in the file • View all records of the file • Search data from the file • Sort the file • Modify data in the file • Delete a record	4	3
6	a) Write a shell script to print the reverse of an input number (do not use REV command) b) Write a shell script to validate the following inputs I. Email ID	2	3

	II. URL c) Write a shell script to concatenate all the given files into a single file.		
7	Write a program to implement the following Unix command in C language using command line arguments a) cp b) cat	2	3
8	Write a C program to implement the CPU scheduling algorithm (FCFS and SPN). Take the number of processes as the user input, along with the arrival time and service time for each process. Calculate the Turn Around Time(TAT) and Waiting Time(WT) for each process. Also, calculate the average TAT and WT.	2	3
9	Write a program in C or any language of your choice that simulates Banker's algorithm for detecting the process's safe state or unsafe state.	4	3
10	Write a C program to simulate page replacement algorithms (a) FIFO (b) LRU Note: Read input from the file and make it dynamic.	2	3

Appendix - 1

Linux Commands

who	who am i	touch	cat
cp	rm	mv	ls
ln	chmod	umask	pwd
mkdir	rmdir	cd	bc
bc -l	expr	factor	logname
uname	tty	date	df
du	ulimit	cal	wc
sort	cut	grep	awk
head	pg	more	tail
pipe ()	tee	ps	kill
nice	read	echo	I/O redirection
top	Bg	fg	jobs

- **Shell programming**

- Shell variables
- Arithmetic in shell script
- Decision-making instructions
- File test
- String test
- Numerical test
- Logical operators
- Case-control structures
- Loop control structures